DATABASE MANAGEMENT SYSTEMS

Model Question Sets

MODEL QUESTION SET 1

Part A: Short Answer / Conceptual Questions

1. What is a file processing system? Mention any two disadvantages.
2. Define DBMS.
3. Differentiate between data and metadata.
4. What is a Data Dictionary?
5. Define data abstraction.
6. What is data independence? Name its types.
7. Define entity and entity set with an example.
8. What is a weak entity set?
9. Define cardinality ratio in an ER model.
10. What is participation constraint?
11. Define relation and relation schema.
12. What is the degree of a relation?
13. Define super key and candidate key.
14. What is a foreign key?
15. Differentiate between relation intension and relation extension.
16. What is RAID?
17. What is an index in a database?
18. What is a B+ Tree?

Part B: Descriptive and Analytical Questions

1. Explain the disadvantages of a traditional file processing system. How does a DBMS overcome these problems?
2. Explain the three levels of data abstraction: physical level, logical level and view level.
3. Explain logical and physical data independence with suitable examples.
4. Draw and explain the system architecture of a DBMS.
5. Explain entity, attribute, relationship, role, cardinality ratio and participation constraint with suitable examples.
6. Differentiate between strong entity and weak entity, and between total participation and partial participation.

7. Consider STUDENT(StudentID, RollNo, Name, Department, Email). Assume StudentID, RollNo and Email are individually unique. Find super keys, candidate keys, the primary key if StudentID is selected, and alternate keys.

8. Consider ORDER_ITEM(OrderID, ProductID, Quantity, UnitPrice). Assuming a product occurs only once in a particular order, identify the candidate key and possible primary key. Explain why neither OrderID nor ProductID alone can be the primary key.

9. Explain entity integrity and referential integrity constraints with examples.

10. Explain the basic organization of magnetic disk storage in a DBMS.

Part C: Long Answer / Problem Solving Questions

1. Design an ER diagram for a College Database. A department has many students and offers many courses. A faculty member belongs to a department. Students register for several courses and a course can be taken by several students. Store the grade obtained by each student in each course. Show entities, attributes, primary keys, relationships, cardinality ratios and participation constraints.

2. Convert the following ER structure into relational schemas: STUDENT(StudentID, Name, Email), COURSE(CourseID, CourseName, Credits), and an M:N relationship ENROLLMENT with Semester and Grade. Identify primary and foreign keys in the resulting relations.

3. Explain B Tree and B+ Tree indexing. Compare them and justify why B+ Tree is commonly used for database indexing.

MODEL QUESTION SET 2

Part A: Short Answer / Conceptual Questions

1. Define database.
2. What is a database system?
3. What is a DBMS?
4. What is metadata? Give one example.
5. What is a schema?
6. Define data model.
7. Name four types of data models.
8. What is a derived attribute?
9. What is a multivalued attribute?
10. What is a relationship set?
11. Define role in an ER relationship.
12. What is an EER model?
13. Define tuple.
14. Define domain.
15. What is relation cardinality?
16. What is a composite key?
17. Define referential integrity.
18. What is ordered indexing?

Part B: Descriptive and Analytical Questions

1. Explain the need for a DBMS using a university database as an example.
2. Differentiate among Database, Database System and DBMS.
3. Explain simple, composite, single-valued, multivalued, stored, derived and key attributes with examples.
4. Explain 1:1, 1:N and M:N relationships with suitable real-world examples.
5. Explain EER features including specialization, generalization and inheritance.
6. Consider EMPLOYEE(EmpID, AadhaarNo, Name, Email, Salary). Assume EmpID, AadhaarNo and Email are individually unique. Determine candidate keys, one possible primary key, alternate keys and at least six super keys.
7. Consider PRODUCT(ProductID, SKU, Barcode, Name, CategoryID). Assume ProductID, SKU and Barcode are unique. Identify super keys, candidate keys, primary key, alternate keys and foreign key, assuming CategoryID references CATEGORY.
8. Explain all major relational model constraints.
9. Explain file organization and record organization in database systems.
10. What is RAID? Explain its purpose in database storage.

Part C: Long Answer / Problem Solving Questions

1. Design an ER diagram for a Hospital Management System. Each patient has PatientID, Name, DOB, Phone and Address. Each doctor has DoctorID, Name and Specialization. A patient can consult many doctors and a doctor can

consult many patients. Consultation has Date, Diagnosis and Fee. A patient may be admitted to a room. Each room has RoomNo and RoomType. Indicate all keys, relationships, cardinalities and participation constraints.

2. Given DEPARTMENT(DeptID, DeptName), EMPLOYEE(EmpID, EmpName, Salary, DeptID), PROJECT(ProjectID, ProjectName) and WORKS_ON(EmpID, ProjectID, Hours), identify primary keys, foreign keys, composite keys and referential integrity constraints.

3. Explain the working of B+ Tree insertion using a suitable example.

4. Construct a neat ER diagram using the following information.

No-Town Recording studio has decided maintain database about its musicians, albums and instruments as follows.Each musician has an unique id, a name, an address, and a phone number, where no address has more than one phone number. Each instrument that is used in songs has a name and a musical key.Each album that is recorded has a title, a copyright date, a format (e.g., CD or MC), and an album identifier. Each song recorded at No-town studio has a title and an author. Each musician may play several instruments, and a given instrument may be played by several musicians. Some instruments are there which are not played by any musician. Each album can contain multiple songs along with each album must contain at least one song, and each song must belong to exactly one album.Each song is performed by one or more musicians, and a musician may perform a number of songs where some musicians might not perform any songs . Each album is produced by exactly one musician. A musician may produce zero or more albums.

5. Construct a neat ER diagram for the hospital database using the following information:

A hospital has multiple specialized wards each identified by a unique name and tracking the total count of current patients. A patient has a unique ID, name, address (place, city, pin), and phone numbers. Patients undergo multiple tests, and each test has a unique test number, category, and final result. Not all patients are advised for tests. A Ward hosts multiple patients. The admission date and time are recorded upon entry. Each ward must have at least one patient. A doctor specialized to/ belongs to one specific ward but each ward can have multiple doctors.Not all wards have specialized doctors. Each patient is assigned to one specific doctor/leading consultants, but other doctors may examine them if needed. Only some doctors are acting as leading consultant for the patients.

MODEL QUESTION SET 3

Part A: Short Answer / Conceptual Questions

1. Why is data redundancy undesirable?
2. What is data inconsistency?
3. What is concurrent access?
4. What is database security?
5. What is physical data independence?
6. What is logical data independence?
7. What is a relational data model?
8. What is an object-oriented data model?
9. What is an object-relational model?
10. What is a recursive relationship?
11. What is an identifying relationship?
12. What is partial participation?
13. What is a relation instance?
14. Can a primary key consist of more than one attribute? Explain.
15. Can a table have more than one candidate key?
16. Can a foreign key contain duplicate values?
17. What is a schema diagram?
18. Why are indices used?
19. Explain the significance of descriptive attributes and NULL attributes in a database with suitable example.

20. Write the fundamental differences between Network and hierarchical model.
21. Mention the basic difference between physical and logical data independence. Draw a labelled diagram of three-schema architecture of database system
22. List any two steps for mapping a ER model into a relational model with suitable example.
23. Define the following ER component: i) Weak and strong entity Set ii) Generalization and Specialization

2. Answer any one. [5x2=10]

3. Answer any one [5x2=10]

(a) List out various integrity constraints in Database Management system.

(b) Discuss about different types of keys in a database system with appropriate example of each.

1. A university stores student, examination and fee data in separate files. Discuss any five problems that may arise and explain how a DBMS can solve them.
2. Explain DBMS architecture with a suitable diagram.

3. Compare ER Model, Relational Model, Object-Oriented Model and Object-Relational Model with examples.

4. Consider EMPLOYEE works_for DEPARTMENT. Each employee works for exactly one department and one department can have many employees. Identify cardinality ratio, participation of EMPLOYEE, possible participation of DEPARTMENT, primary keys and foreign key after relational mapping.

5. Consider CUSTOMER(CustomerID, AadhaarNo, Email, Name, City). Assume CustomerID, AadhaarNo and Email uniquely identify customers. If CustomerID is selected as the primary key, find all candidate keys, alternate keys, five super keys and explain why {CustomerID, Name} is a super key but not a candidate key.

6. Consider STUDENT(StudentID, Name, DeptID) and DEPARTMENT(DeptID, DeptName). Identify primary and foreign keys and explain the referential integrity constraint.

7. Explain ER to relational mapping for strong entity, weak entity, 1:1, 1:N and M:N relationships.

8. Explain sequential, indexed and hashed file organization.

9. Differentiate between primary index, secondary index and clustering index.

10. Compare B Tree and B+ Tree.

Additional Practice: Finding Keys in Relations

Problem 1: STUDENT(StudentID, RollNo, Name, Email, Department)

StudentID, RollNo and Email are individually unique. Find candidate keys, primary key, alternate keys and examples of super keys.

Problem 2: EMPLOYEE(EmpID, AadhaarNo, Email, Name, Department)

EmpID, AadhaarNo and Email are individually unique. Find all candidate keys and suitable alternate keys.

Problem 3: BOOK(BookID, ISBN, Title, Author, Publisher)

BookID and ISBN are unique. Find candidate keys and at least eight super keys.

Problem 4: PRODUCT(ProductID, SKU, Barcode, Name, CategoryID)

ProductID, SKU and Barcode are unique. Find candidate keys, primary key and alternate keys.

Problem 5: ORDER_ITEM(OrderID, ProductID, Quantity, UnitPrice)

The same product occurs at most once in an order. Find the minimal key and justify why it is composite.

Problem 6: ENROLLMENT(StudentID, CourseID, Semester, Grade)

A student may register for many courses and a course may have many students. Identify a suitable composite primary key.

Problem 7: FLIGHT(FlightNo, FlightDate, Source, Destination, AircraftNo)

The same FlightNo can operate on different dates. Determine an appropriate candidate key.

Problem 8: ROOM_BOOKING(RoomNo, BookingDate, TimeSlot, CustomerID)

A room may be booked multiple times on the same day but not for the same time slot. Find the candidate key.

Problem 9: RESULT(StudentID, SubjectCode, Semester, Marks, Grade)

A student has one result for each subject in a particular semester. Find an appropriate composite key.

Problem 10: BANK_ACCOUNT(AccountNo, CustomerID, BranchCode, AccountType, Balance)

AccountNo uniquely identifies every account. Identify candidate key, primary key, possible foreign keys and examples of super keys.

1. Can every super key be a candidate key? Justify with an example.
2. Can two candidate keys exist in the same relation? If yes, which one becomes the primary key?
3. Why is {OrderID, ProductID} a candidate key in ORDER_ITEM while OrderID alone is not?
4. Why can Email be an alternate key even though it uniquely identifies a customer?
5. Can a foreign key be NULL? Explain with an appropriate relationship.
6. Can a foreign key contain duplicate values? Justify using EMPLOYEE and DEPARTMENT.
7. Why must a candidate key be minimal while a super key need not be minimal?
8. Can a composite primary key contain two foreign keys? Explain using ENROLLMENT or ORDER_ITEM.
9. What happens if a foreign key value does not exist in the referenced relation? Which constraint is violated?
10. Why is B+ Tree generally preferred to B Tree for database indexing?